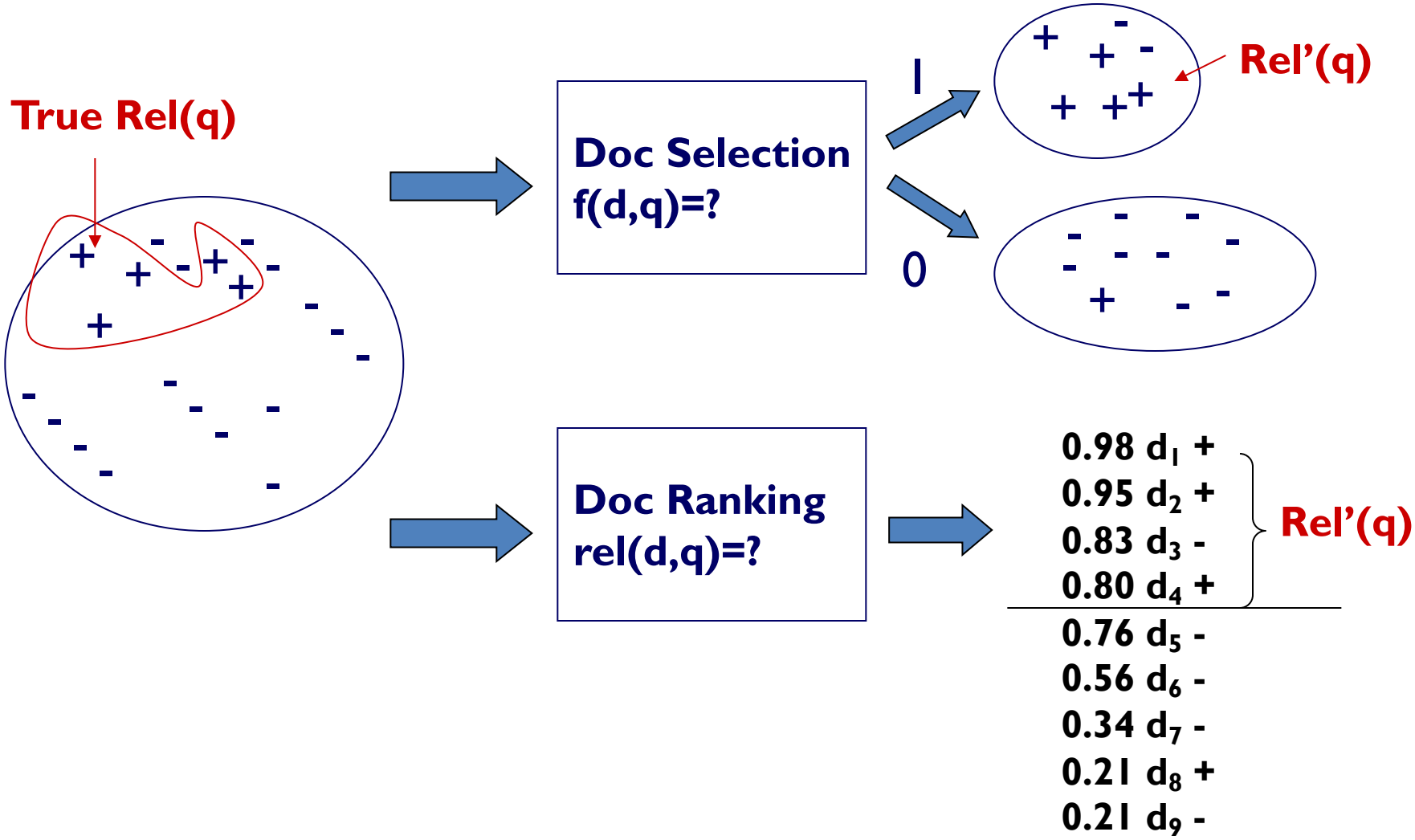
Vector Space Model

Some slides borrowed from Hongning Wang with modifications

Document Selection vs. Ranking



Intuitions for Ranking

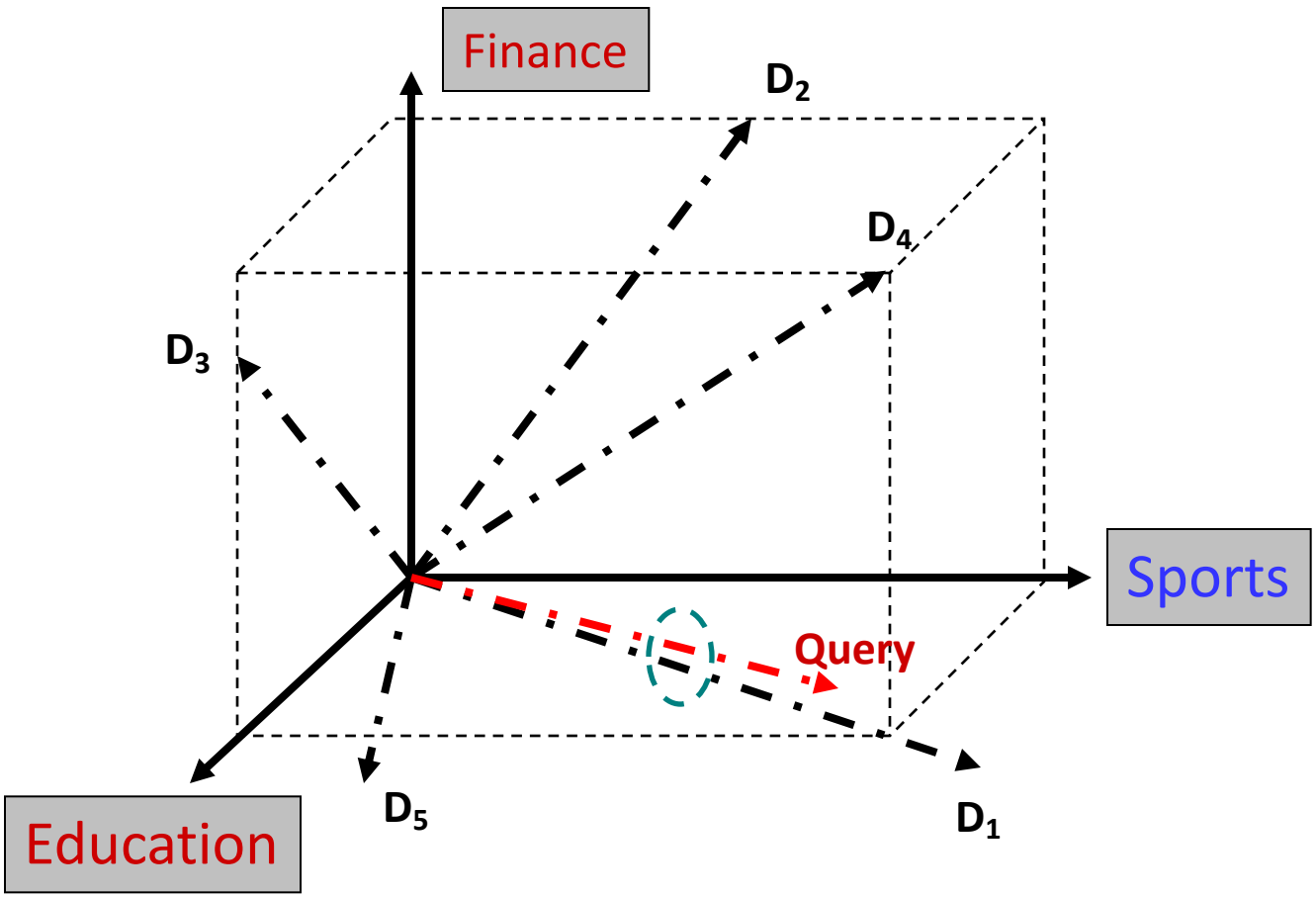
- Query side: some terms are more important than others to represent the user's information need
- Document side: some terms carry more information about the document

Vector space model

- Represent both document and query by <u>concept</u> vectors
  - Each concept defines one dimension
  - *K* concepts define a high-dimensional space
  - Element of vector corresponds to concept weight
    - E.g., $d=(x_1,\ldots,x_k)$, x_i is "importance" of concept i
- Measure relevance
  - Similarity between the query vector and document vector in this concept space

VS Model: an illustration

- Which document is closer to the query?



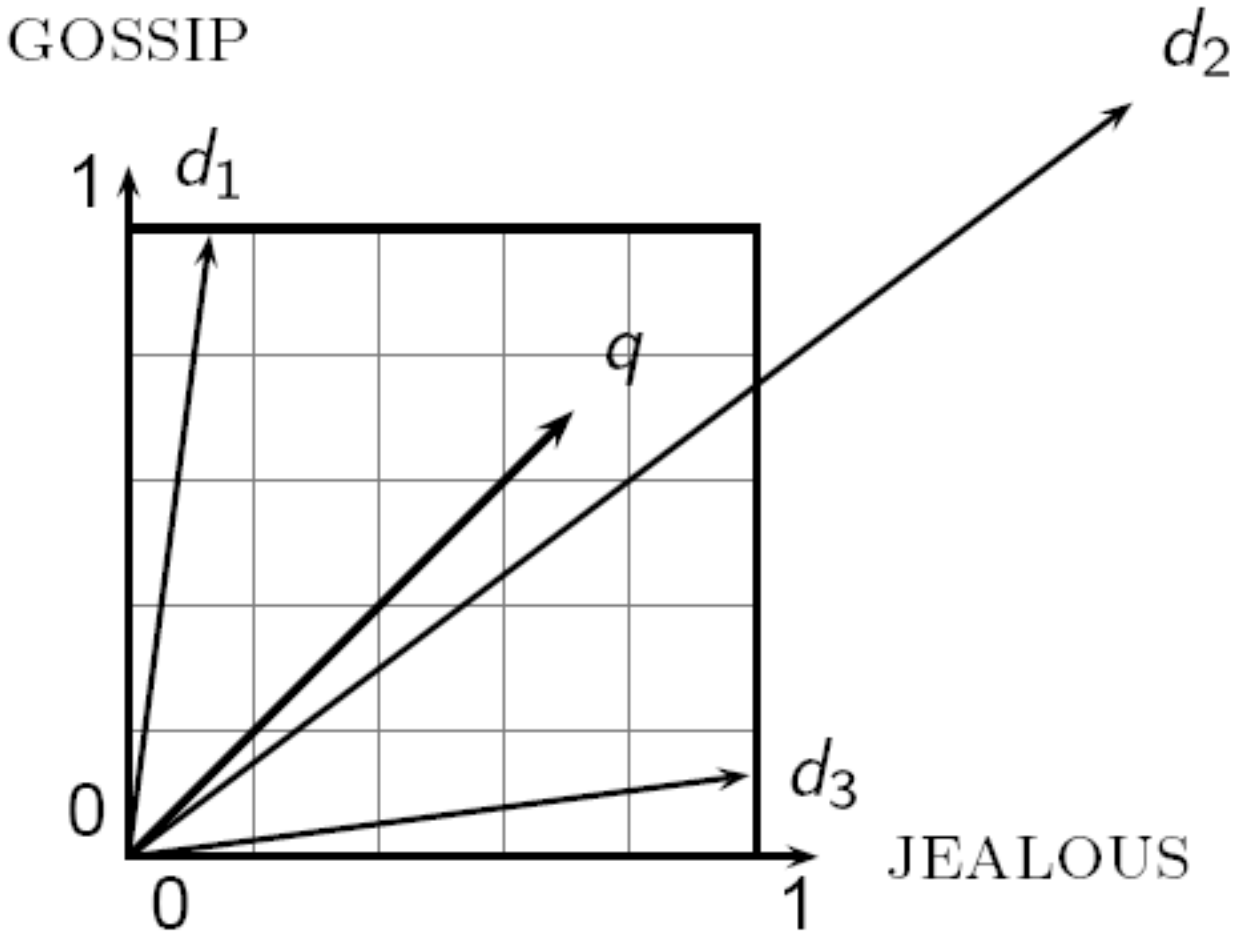
What the VS model doesn't say

- How to define/select the "basic concept"
  - Concepts are assumed to be <u>orthogonal</u>
- How to assign weights
  - Weight in query indicates importance of the concept
  - Weight in doc indicates how well the concept characterizes the doc
- How to define the similarity/distance measure

What is a good “basic concept”?

- Orthogonal
  - Linearly independent basis vectors
    - “Non-overlapping” in meaning
    - No ambiguity
- Weights can be assigned automatically and accurately
- Existing solutions
  - Terms or N-grams, i.e., bag-of-words
  - Topics, i.e., topic model

Bag of words representation



How to assign weights?

- Important!
- How?
  - Two basic heuristics
    - TF (Term Frequency) = Within-doc-frequency
    - IDF (Inverse Document Frequency)

TF weighting

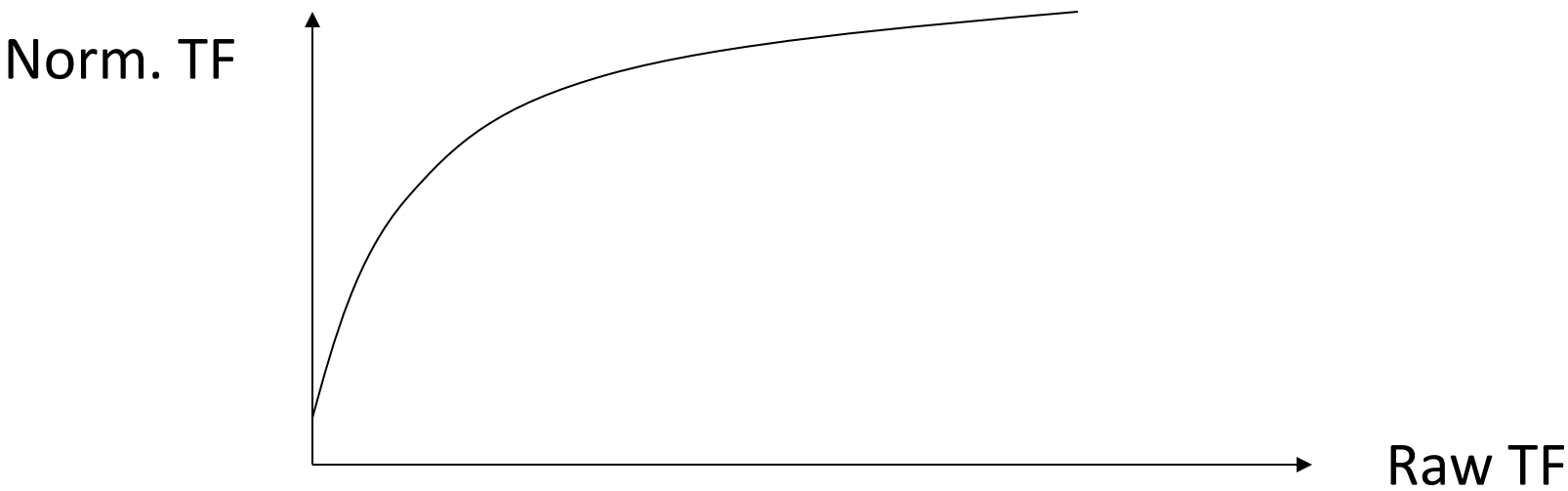
- Idea: a term is more important if it occurs more frequently in a document
- TF Formulas
  - Let $f(t, d)$ be the frequency count of term t in doc d
  - Raw TF: $tf(t, d) = f(t, d)$

TF normalization

- Query: *iphone* 15 Pro Max
  - D1: iPhone 15 Pro Max launched in September of last year.
  - D2: iPhone 15 Pro Max has four color options.
  - D3: iPhone 15 Pro Max has four color options. iPhone 15 Pro Max has four color options. iPhone 15 Pro Max has four color options.

TF normalization

- Sublinear TF scaling
  - $tf(t,d) = \begin{cases} 1 + \log f(t,d)\,, if\ f(t,d) > 0 \\ 0, otherwise \end{cases}$



Document frequency

- Idea: a term is more discriminative if it occurs only in fewer documents

IDF weighting

- Solution
  - Assign higher weights to the rare terms
  - Formula
    - $IDF(t) = \log(\frac{N}{df(t)})$

Non-linear scaling

Total number of docs in collection

Number of docs containing term t

  - A corpus-specific property
    - Independent of a single document

Collection vs. Document frequency

- Collection frequency of *t* is the total number of occurrences of *t* in the collection (incl. multiples)
- Document frequency is number of docs *t* is in
- Example:

Word	Collection frequency	Document frequency
insurance	10440	3997
try	10422	8760

- Which word is a better search term (and should get a higher weight)?

tf-idf weighting has many variants

Term frequency		Document frequency		Normalization	
n (natural)	$\mathrm{tf}_{t,d}$	n (no)	1	n (none)	1
l (logarithm)	$1+\log(\mathrm{tf}_{t,d})$	t (idf)	$\log \frac{N}{\mathrm{df}_t}$	c (cosine)	$\frac{1}{\sqrt{w_1^2+w_2^2+\ldots+w_M^2}}$
a (augmented)	$0.5+\frac{0.5\times\mathrm{tf}_{t,d}}{\max_t(\mathrm{tf}_{t,d})}$	p (prob idf)	$\max\{0,\log\frac{N-\mathrm{df}_t}{\mathrm{df}_t}\}$	u (pivoted unique)	$1/u$
b (boolean)	$\begin{cases}1 & \text{if } \mathrm{tf}_{t,d}>0\\ 0 & \text{otherwise}\end{cases}$			b (byte size)	$1/CharLength^{\alpha}$, $\alpha<1$
L (log ave)	$\frac{1+\log(\mathrm{tf}_{t,d})}{1+\log(\mathrm{ave}_{t\in d}(\mathrm{tf}_{t,d}))}$				

TF-IDF weighting

- Combining TF and IDF
  - Common in doc → high tf → high weight
  - Rare in collection→ high idf→ high weight
  - $w(t,d) = TF(t,d) \times IDF(t)$
- Most well-known document representation schema in IR! (G Salton et al. 1975, A vector space model for automatic indexing)

"Salton was perhaps the leading computer scientist working in the field of information retrieval during his time." - wikipedia

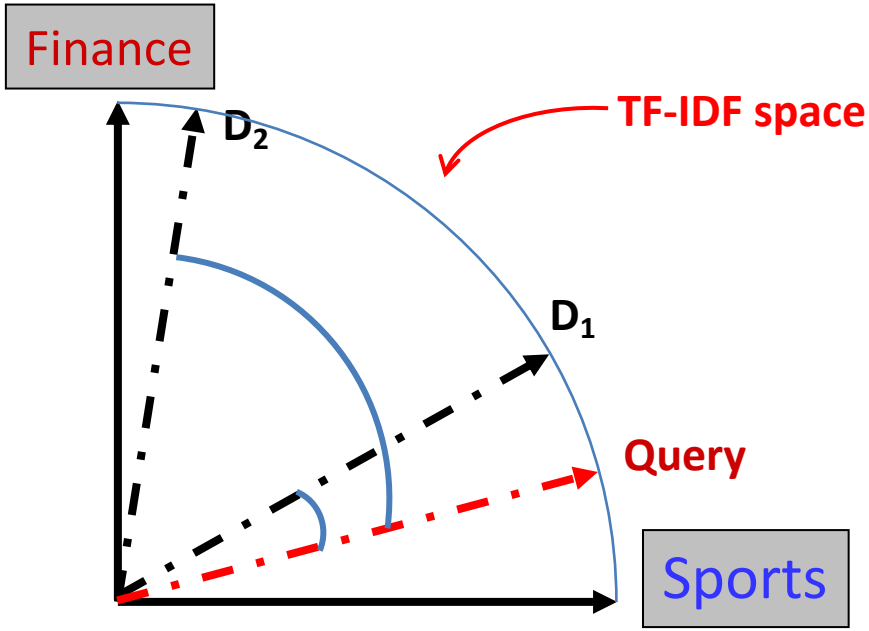
Gerard Salton Award – highest achievement award in IR

Cosine similarity

- Angle between two vectors
  - $cosine(V_q, V_d) = \frac{V_q \times V_d}{|V_q|_2 \times |V_d|_2} = \frac{V_q}{|V_q|_2} \times \frac{V_d}{|V_d|_2}$
  - Document length normalized

TF-IDF vector

Unit vector



What you should know

- Basic idea of vector space model
- Two important heuristics in VS model
  - TF
  - IDF
- Similarity measure for VS model
  - cosine similarity

Today's reading

- Chapter 6: Scoring, term weighting and the vector space model
  - 6.2 Term frequency and weighting
  - 6.3 The vector space model for scoring
  - 6.4 Variant tf-idf functions

An Example

Consider the following two documents:

D1: Texas forever! Texas forever! Coach believes Texas forever!
D2: Coach believes Texas Aggies win! Aggies win! Aggies win!

In this problem, we are going to apply our skills at vector space retrieval to determine the similarity between the two documents.

a) First, we need to pre-process the documents. Remove all punctuation and perform casefolding (convert to all lower cases). Now write down the modified documents:

b) Now, we're going to view our documents in terms of the vector space model. How many dimensions (axes) are there in the vector space corresponding to our two documents?

c) Now write down the two vectors that correspond to the documents assuming simple term frequency weights (no IDF factor, no logarithms).

d) What is the length (magnitude) of each vector?

e) Finally, we're ready to determine the similarity between document 1 and document 2. What is the cosine similarity between document 1 and document 2?

The Answer

Part a

D1: texas forever texas forever coach believes texas forever

D2: coach believes texas aggies win aggies win aggies win

Part b

6 dimensions.

Part c

	texas	forever	coach	believes	aggies	win
D1	3	3	1	1	0	0
D2	1	0	1	1	3	3

Part d

D1: sqrt(9 + 9 + 1 + 1) = sqrt(20)

D2: sqrt(1 + 1 + 1 + 9 + 9) = sqrt(21)

Part e

Cosine Similarity = (3 + 1 + 1) / [sqrt(20)*sqrt(21)] = 5 / sqrt(420)